

Cracking lab

Revise 360 lesson plan — 60 minutes

Unit	OCR J277 1.4 Network security — Paper 1, Computer systems
Lesson	Cracking lab (lesson 6 of the unit)
Lesson ID	ns-l06
Experience ID	ns-l06
Plan ID	PLAN-NS-L06
Content version	b51613070dc8
Website	https://revise360.co.uk/experience.html?id=ns-l06

Big question, outcomes and success criteria

Big question — How does a brute-force attack work?

Prerequisites

The lesson before this one is ns-l05. Pupils should already have finished it: the starter below retrieves from it.

Earlier in the unit: ns-l01, ns-l02, ns-l03, ns-l04.

Assessable outcomes

1. **ns-l06-o1** — understand what a brute-force attack is
2. **ns-l06-o2** — understand how to identify and protect against brute-force attacks

Vocabulary, resources and preparation

Key vocabulary — Brute-force attack, Password

Resources

SLIDES	presentations/NS_L06_CrackingLab_Lesson.pptx
WORKSHEET	worksheets/NS_L06_CrackingLab_Worksheet.docx
ONLINE EXPERIENCE	experience.html?id=ns-l06

Before the lesson

1. Sign in on the **teacher dashboard** (teacher.html) and open **Class logins**. Enter the class or group and the usernames, one per line, then print the login cards. A pupil cannot sign in without one, so do this before the lesson rather than in it.
2. Print the worksheet, one per pupil.
3. Open the slides and the experience once yourself, on the machine you will be teaching from, so the 360 image is already cached.
4. Check headsets or devices are charged and on the school network.

How pupils get in

1. Go to topics.html on revise360.co.uk.
2. Sign in with the **username and PIN from the login card**. The class is already set against the card.
3. Choose the topic, then this lesson's card.

Worth knowing before you promise anything. In this build `APP_CONFIG.backendUrl` is empty, so logins, login cards and progress are stored on the device the pupil used. The teacher dashboard says so too. A pupil who moves to a different machine starts again, so keep a class on the same devices within a unit, and treat the paper worksheet as the record that travels.

Teaching sequence

Sixty minutes including transitions. The online block is the one to move if your lesson is shorter: see the adaptation at the end.

MIN	PHASE	TEACHER	PUPILS	EVIDENCE TO CHECK
0-5	Retrieve	Starter on the board or the worksheet, from the last lesson and this lesson's prerequisites.	Answer the starter on the worksheet.	The starter box on NS_LO6_CrackingLab_Worksheet.docx is filled in.
5-8	Review the starter	Take answers; correct the one most got wrong.	Correct their own starter.	A correction written, not just a tick.
8-18	Explain and model	The new idea in two or three steps, with the example worked in front of the class.	Listen and answer questions about the example.	They can say why, not only what.
18-21	Set up and settle	Sign-in, launch, and what to write before anyone taps an answer.	Sign in, open the lesson, find What is a brute-force attack?.	Everyone is in and on station 1.
21-43	Online: work the stations in order	Pupils read each wall, write the key fact and the challenge, then answer at the badge.	Read each wall, write the key fact and the challenge, then answer at the badge.	The worksheet has writing on it for each of the 6 stations before the badge is tapped.
43-48	Check understanding	The hinge question, hands down.	Answer the hinge question, hands down.	The split across the options.
48-56	Apply	Exam practice on the worksheet, in silence.	Exam practice, in silence.	worksheet, Exam practice a) to c)
56-60	Exit and set up the next lesson	The exit question and the retrieval link to next lesson.	Write the exit answer.	One sentence that answers the big question.

The explanation and the worked example

This is what the walls of the experience say, station by station. Teach the first two or three from the front before anyone puts a headset on; the rest the pupils read for themselves.

1. What is a brute-force attack?

- Software tries every possible combination of characters until it finds the password.
- No clever trick is needed: just speed, and modern computers are very fast.
- It's often automated and aimed at thousands of accounts at once.

Challenge on this wall: Why is a brute-force attack described as 'trial and error'?

2. Dictionary attacks

- A dictionary attack tries lists of common passwords and real words first.
- Lists from previous data breaches are used, because people reuse passwords.
- 'password', '123456' and 'qwerty' are tried within the first second.

Challenge on this wall: Why is 'Liverpool2015' a weaker password than it looks?

3. How long would it take?

- Each extra character multiplies the number of possible passwords.
- Using upper case, numbers and symbols increases the choices per character.
- A short password can fall in seconds; a long passphrase could take centuries.

Challenge on this wall: Use the starter site to test a password you don't use. How could you improve it?

4. Password policies

- Organisations set rules: a minimum length, a mix of character types, and no reuse.
- Passphrases of three random words are long, strong and memorable.
- Password managers create and store a different strong password for every site.

Challenge on this wall: Write three rules for a school password policy, and explain each.

5. Fighting back

- Account lockouts: after a few failed attempts the account is locked or delayed.
- CAPTCHA tests block automated software.
- Two-factor authentication means a stolen password isn't enough.
- Rate limiting slows down repeated attempts from the same source.

Challenge on this wall: Why does a delay of a few seconds after a failed login make brute force much harder?

6. Behind the scenes

- Websites should store a hash of your password, not the password itself.
- Hashing is one-way: you can check a password, but not work it out from the hash.
- A 'salt' adds random data so identical passwords have different hashes.

Challenge on this wall: Why is it serious if a website stores passwords as plain text?

Retrieval and the check for understanding

Starter — Estimate how many guesses a computer would need to find a 4-digit PIN, then a 6-letter lower-case password. What does that tell you about password length?

Answer space: 3 lines on the worksheet.

The answer is in the teacher answers for this unit, not on this plan: a plan left on a desk should not give it away.

Hinge question

Use this one, hands down, before anyone moves on to independent work. It is a real activity from the experience (ns-106-s3-c2), so the wording a pupil sees matches the wording you say.

Why does adding one character make such a difference?

Correct: Every extra character multiplies the number of possible combinations

What a pupil might choose instead

- It confuses the attacker
- It makes the password encrypted
- It blocks the login page

Count the hands on each. If more than a few of the class are on one wrong option, that option is the misconception to teach against before anyone goes on: go back to the station this question sits on and work through its example again, aloud, before the independent work.

The course's own response to a wrong answer here: With 95 possible characters, each extra one multiplies the possibilities by 95.

One response, not four. one response covers every wrong option; it does not say what to revisit for each one, so decide your own reteach for each option before the lesson - the list above is the start of it.

Guided to independent practice

STAGE	ACTIVITIES	WHAT IT IS
Recognise	12	The answer is on the screen and the pupil picks it: multiple choice, choosing several, matching a pair.
Construct	1	Nothing is offered. The pupil classifies, orders, fills in a table or works a number out, and types or places the answer.

Scaffolds already in the product, and how they fade

- **Hint** is a ladder, not a door: the first rung names the idea, the next shows the shape of the code, the next how it starts, the last animates the technique on different data. A pupil asks for each rung. None of them is the answer.
- **Run** is free and unmarked; **Check my answer** marks. A pupil may run as often as they like, which is what makes trial and error safe.
- **I need help** says to ask you, and says plainly that the question still has to be completed. It does not notify anyone and it does not unlock anything.
- The worked example appears on every early activity and on none of the late ones. That fade is deliberate; it is not a missing example.

Accessibility

- **Read aloud** on every task window reads the task and its brief, with the prescribed values read as values rather than as punctuation.
- Text is 16px or larger throughout, and a question window scrolls rather than shrinking its type.
- Prescribed data is boxed and in a monospace face as well as coloured, so it still reads as code in black and white, in a high-contrast mode, and to a pupil who does not separate green from amber.
- **Lower literacy and EAL:** pre-teach the vocabulary above; let the pupil say the key fact before writing it; accept the key fact in two or three words.
- **SEND:** the sequence is compulsory but the pace is not. A pupil may finish a station next lesson; the progress saves after every answer and reopens at their earliest unfinished question.
- A headset is never required. Every experience works on a laptop or a tablet with a mouse or a finger.

Delivering the online experience

Before anyone enters

- Worksheet out, pen in hand. The rule is write first, then answer at the badge: the first answer is the one that counts.
- Say where the stations are. They are numbered around the room: 1 and 2 to the right, 3 and 4 behind, 5 and 6 to the left, and the final challenge on the floor.
- Say that a badge marked **Not yet** is not a fault. It opens when the station they are on is finished.

The stations, and what to record at each

#	STATION	ACTIVITIES	RECORD ON PAPER
1	What is a brute-force attack?	2	A brute-force attack is...
2	Dictionary attacks	2	A dictionary attack works by...
3	How long would it take?	2	Password strength depends on...
4	Password policies	2	A good password policy says...
5	Fighting back	2	Login pages defend against brute force by...
6	Behind the scenes	2	Websites should store passwords by...
final	Final challenge: password strength	1	the answer to the station's challenge

Where to pause

Stop the class after station 2 and take answers to its challenge out loud. That is the point where a pupil who has misread the technique can still be put right cheaply.

Afterwards

Headsets off and down. Ask three pupils for the key fact from a station of your choosing, in their own words and not read off the sheet. Then the exit question.

Assessment, evidence and unfinished work

ID	QUESTION	MARKS	WHERE
ns-106-x1	Describe how a brute-force attack works.	2	worksheet, Exam practice a); deck, Exam practice slide
ns-106-x2	Explain two measures a website can use to make brute-force attacks impractical.	4	worksheet, Exam practice b); deck, Exam practice slide
ns-106-x3	Explain why a long passphrase is harder to crack than a short complex password.	3	worksheet, Exam practice c); deck, Exam practice slide

Expected answers

Model answers and marking guidance are teacher-only and are not printed on this plan. They are in the unit's teacher guide pack, and the per-activity marking for this lesson is in `answers/records/ns-106.json`, which is not published with the website.

Common errors

The wrong options the activities offer are the errors this lesson expects. The commonest to watch for:

- **What is a brute-force attack?** — Tricking a user into revealing a password; Flooding a server with traffic
- **Which account is most at risk from brute force?** — One with a long passphrase; One that uses two-factor authentication
- **How is a dictionary attack different from pure brute force?** — It only works on dictionaries; It attacks the router
- **Why is reusing a password across sites dangerous?** — It uses more storage; It slows down logins
- **Which password is hardest to brute-force?** — P@ss1; Summer2024

Unfinished work

Record where each pupil stopped, not a tick. They resume at their earliest unfinished question, and a station left half-done is the thing to open next lesson. Do not mark the lesson complete for a pupil who did not finish it: the score screen already shows what was and was not done, and that is the honest record.

Exit, next lesson and homework

Exit check

One of these, written, not a show of hands:

- What a brute-force attack is
- What makes a password strong
- How login pages defend against brute force

The confidence columns on the worksheet are a supplement to this, not a substitute: a pupil who rates themselves confident and cannot answer the exit question has told you nothing.

Retrieval link to next lesson

Next is ns-107, Traffic flood. Use this lesson's big question as next lesson's starter, so the first thing they do is retrieve it.

Homework

Finish any station left open, on the website, from the question they stopped at. If the class has no devices at home, set the exam practice on the worksheet instead and mark it next lesson.

Shorter lessons, longer lessons and no internet

A 40 to 45 minute lesson

Keep the starter, the explanation and the model at full length: cutting those is what makes the online work fail. Take the time out of the online block and the apply phase. Pupils stop where they are; their progress saves after every answer and reopens at their earliest unfinished question, so the next lesson starts there.

If the internet or the devices are unavailable

Teach from the slides and the worksheet. The slides carry the same teaching as the station walls, and the worksheet carries the key facts, the challenges and the exam practice. Set the online stations when the devices are back.

There is no offline version of the experience, and this plan does not contain one. Do not hand out a printed walkthrough of the stations as a substitute: the experiences are used on the website.

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